

From World Models to Noticing Machines

Affective State, Predictive Control, and Symbolic Governance in Embodied AI Systems**

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Abstract

Recent advances in artificial intelligence have emphasized world models—systems that learn latent representations of environments and predict future states—as a path toward planning and generalization. However, predictive capability alone does not determine which predictions matter, when action should be suppressed, or how irreversible decisions should be governed. In biological systems, these functions are mediated by affective and regulatory processes that operate prior to action.

In this work, we introduce the Noticing & Tethering Core (N&T Core), an intermediate architectural layer that converts multimodal world-state representations into affect-weighted control signals. Rather than optimizing reward or memory retention, the N&T Core regulates salience, recursion depth, and action eligibility through a structured affective state space. We further describe a symbolic governance layer that enforces constraints on irreversible actions using auditable, zero-knowledge-compatible decision gates.

We show how the integration of world models, affective control, and symbolic governance enables predictive systems to operate safely in embodied, social environments without fine-tuning, persistent identity tracking, or raw data retention. This architecture reframes emotion not as sentiment or cognition, but as a computational control surface for prediction and planning.

1. Introduction

World models have emerged as a promising framework for artificial intelligence systems capable of planning, abstraction, and generalization. By learning latent state representations and modeling their temporal evolution, such systems can predict the consequences of actions across time horizons. Recent work has emphasized the importance of grounding these models in sensory data—particularly vision and embodied interaction—to overcome the limitations of purely text-trained systems.

Despite these advances, world models face a critical unresolved problem: prediction alone does not imply regulation. A system may accurately forecast future states while lacking a principled mechanism to determine which states warrant intervention, which actions should be delayed, and which outcomes are too irreversible to execute under uncertainty.

In human and animal cognition, this regulatory function is not performed by explicit reasoning or memory recall alone. Instead, affective processes—often labeled “emotion”—weight perception, suppress or amplify responses, and gate behavior under uncertainty. Importantly, these processes operate before deliberate action, shaping decision boundaries rather than outcomes.

Current AI architectures typically address regulation through external mechanisms such as reward functions, policy constraints, reinforcement learning fine-tuning, or post hoc safety filters. These approaches, while effective in constrained domains, struggle in open-ended, multimodal, and socially embedded environments where context shifts rapidly and irreversible actions carry asymmetric risk.

This paper proposes a different approach. We introduce an intermediate architectural layer—situated between world-state estimation and action execution—that we call a Noticing Machine. Its core component, the Noticing & Tethering Core (N&T Core), transforms latent world representations into structured affective signals that regulate planning, recursion, and action eligibility. These signals do not encode goals or rewards; instead, they encode importance, instability, and restraint.

We further introduce a symbolic governance layer that enforces constraints on irreversible actions through auditable decision gates. This layer operates without storing raw sensory data or user identity, enabling compliance and oversight without surveillance.

Together, these components address a central gap in current world model architectures: the absence of an internal mechanism for affect-weighted control and lawful action gating. We demonstrate how this approach enables predictive systems to function safely and coherently in embodied social environments, while remaining compatible with existing model architectures and deployment constraints.

2. Background

2.1 World Models and Predictive Architectures

World models aim to learn compact latent representations of an environment and to predict how these representations evolve over time. Typically, such systems consist of three components:

- an encoder that maps observations to a latent state,

- a dynamics model that predicts future latent states, and

- a decoder that maps latent states back to observable or task-relevant quantities.

This paradigm has been applied successfully in robotics, control, and simulation-based reinforcement learning, and has recently been extended to multimodal and video-based systems. The central advantage of world models lies in their ability to decouple perception from planning, enabling prediction without direct interaction.

However, most world model architectures implicitly assume that all predictions are equally actionable. Regulation—when present—is typically handled by downstream policy networks or external reward structures. As a result, the model itself lacks a notion of salience, uncertainty tolerance, or action suppression beyond probabilistic error.

2.2 Limitations of Reward- and Policy-Based Regulation

Reinforcement learning introduces regulation through reward maximization, but this approach conflates value with safety. Reward functions encode desired outcomes, not conditions under which action should be withheld. Moreover, reward shaping often requires extensive task-specific tuning and can fail catastrophically in distributional shifts.

Similarly, policy constraints and rule-based safety layers are typically applied after planning has occurred. These mechanisms operate as filters rather than intrinsic regulators, and they do not provide a graded notion of hesitation, silence, or deferment.

In contrast, biological systems exhibit continuous regulation of perception and action through affective processes that are neither reward signals nor explicit rules. These processes modulate attention, suppress impulsive responses, and introduce hesitation under uncertainty—functions that are largely absent from current predictive AI architectures.

3. The Architecture of a Noticing Machine

We define Noticing as the process by which a system detects that something matters prior to deciding what to do. Noticing is not equivalent to attention, memory, or anomaly detection. Instead, it is a regulatory transformation that maps world-state information into internal signals governing readiness, restraint, and escalation.

3.1 The Noticing & Tethering Core (N&T Core)

The N&T Core operates on inputs derived from a world model’s latent state and predictions. Its output is a structured internal state we call a Feature Vector, composed of multiple regulatory dimensions:

Affective Weighting (“Ache”): a set of scalar projections representing the intensity and importance of detected conditions.

Tone Metrics: measures of coherence, volatility, and cadence in the system’s internal state evolution.

Drift Indicators: signals that quantify instability or divergence across recursive planning cycles.

Silence Slots: explicit markers that suppress action or output when uncertainty exceeds acceptable bounds.

Recursion Depth Controls: limits on iterative planning under escalating uncertainty.

These components do not encode goals or preferences. Instead, they encode constraints on cognition and action.

3.2 Tethering and Action Eligibility

The second function of the N&T Core is Tethering: the generation of directives that bind planning and action to regulatory conditions. Tethering directives may include:

required confirmation thresholds,

maximum allowable recursion depth,

persona or role constraints,

and symbolic constraints on action categories.

Tethering ensures that planning remains anchored to regulatory state, preventing runaway optimization or premature execution.

4. Affective State as a Computational Control Surface

In this architecture, affective signals are treated as control variables, not emotional labels. They modulate system behavior in three primary ways:

Weighting: increasing or decreasing the influence of predicted outcomes.

Suppression: explicitly inhibiting action or output.

Gating: conditioning the legality or reversibility of actions on stability thresholds.

Silence, in this context, is not absence but active regulation. The system may choose not to act, not to respond, or not to escalate—even when prediction confidence is high—if affective signals indicate instability or disproportionate risk.

5. Integration with World Models

When integrated with a world model, the Noticing Machine operates as follows:

The world model produces a latent state and predicted futures.

These are transformed into regulatory signals by the N&T Core.

Planning proceeds only within the bounds defined by tethering directives.

Actions are passed through a symbolic governance gate prior to execution.

This architecture ensures that prediction informs action without compelling it.

6. Symbolic Governance and Irreversible Action Control

Irreversible actions—such as permanent memory writes, external notifications, or legal assertions—require additional safeguards. We introduce a symbolic governance layer that evaluates proposed actions against regulatory state without accessing raw data or identity.

This layer operates on symbolic summaries and produces auditable decisions: allow, defer, or block. Because it relies on symbolic representations rather than stored content, it enables oversight and compliance without surveillance or retention of sensitive information.

7. Case Study: Embodied Social Environment

We apply this architecture to a real-time, video-grounded social environment characterized by rapid context shifts and asymmetric risk. The system maintains predictive awareness of crowd dynamics while suppressing or deferring actions under instability. Notably, no fine-tuning or persistent identity tracking is required.

8. Implications and Future Work

This work suggests that scalable AI safety and governance may depend less on better policies or larger models, and more on the introduction of intermediate regulatory layers. By separating prediction, affective control, and symbolic governance, systems can act with restraint without sacrificing capability.

9. Conclusion

World models provide the ability to predict. Noticing machines provide the ability to refrain. Together, they form a foundation for AI systems that are not merely intelligent, but governable.

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